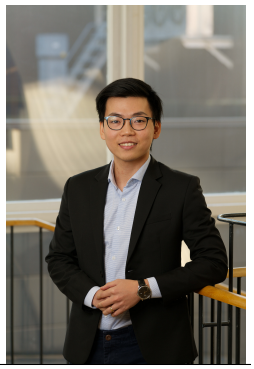
Hung Le

Doctoral Candidate at Aalto University, Finland



Personal Information

- Email: hung.h.le@aalto.fi
- Website: https://hungleeconomics.github.io/

Education

- **Aalto University** — Espoo, Finland
  Doctor of Science in Economics and Business Administration, Economics; GPA: 4.59/5 — *Sep 2022 - 2027 (Expected)*
- **Aalto University** — Espoo, Finland
  Master of Science, Economics, Research Track; GPA: 4.58/5 — *Sep 2019 - Apr 2022*
- **Aalto University** — Mikkeli, Finland
  Bachelor of Science, International Business; GPA: 4.93/5 — *Sep 2016 - Feb 2019*

Job Market Paper

- **Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market** with Matias Pousi
  Full Paper

  Abstract: While price regulation is widely used to decrease prices, it can inadvertently provide firms with a focal point for coordination. The reference price system adopted in the Finnish pharmaceutical market in 2009 initially resulted in lower prices, but after a few years, prices gradually returned to their pre-reform level. How did firms succeed in raising prices, and what role did the regulation play in this process? We find that firms learned to use the previous quarter's reference price as a focal point when choosing prices at the beginning of each quarter. This pattern led to price increases from quarter to quarter. Six big generic firms were more likely to use focal-point pricing, and their presence was associated with similar behavior among smaller rivals. According to our structural model, 65% of products had a profitable downward deviation from the upward pricing path but only 13% deviated. The remaining 52% suggests potential tacit coordination. On the positive side, we show that after the quarterly reference price was fixed at the beginning of the quarter, it constrained firms from increasing prices higher. Furthermore, a regulatory change in 2017 increased price competition by weakening firms' incentives to use focal-point pricing.

Experience

- **Aalto University** — Espoo, Finland
  Teaching Assistant
  - Math Camp, Ph.D. level: Aug 2024, Aug 2025
  - Empirical Industrial Organization, M.Sc. level: Oct 2024, Oct 2025
  - Intermediate Macroeconomics II, B.Sc. level: Mar 2023, Mar 2024, Mar 2025
  - Principles of Economics, B.Sc. level: Sep 2022, Sep 2023, Sep 2024, Sep 2025
  - Mathematics for Economists, B.Sc. level: Oct 2021
- **Aalto University** — Espoo, Finland
  Research Assistant — *Jun 2022 - Aug 2022*
- **VATT Institute for Economic Research (Valtion taloudellinen tutkimuskeskus)** — Helsinki, Finland
  Summer Trainee / Research Assistant — *May 2021 - May 2022*

References

- Professor Otto Toivanen, Aalto University (Main Supervisor)
  otto.toivanen@aalto.fi
- Professor Pauli Murto, Aalto University
  pauli.murto@aalto.fi
- Professor Alina Ozhegova, Aalto University
  alina.ozhegova@aalto.fi

Honors and Awards

- Dean's List, Dec 2019
- Ilkka Kontula Foundation Award, May 2018

Grants Awarded

- OP Research Foundation, Mar 2025, Mar 2026
- Yrjö Jahnsson Foundation, Dec 2023
- Jenny and Antti Wihuri Foundation, Sep 2023
- Foundation for Economic Education Liikesivistysrahasto, May 2022, Nov 2024, Nov 2025

Skills

- Software: Stata, LaTeX, Microsoft 365, Python, R
- Language: English (fluent), Vietnamese (native), Finnish (basic)

Conferences and Seminars

- **Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market (JMP)**
  - European Commission DG-Comp (Brussels, 2026, invited seminar)
  - Finnish Competition and Consumer Authority (online, 2026, invited seminar)
  - Nordic workshop on Industrial Organization NORIO (Copenhagen, 2026, workshop)
  - European Winter Meeting of the Econometric Society (Warsaw, 2026, conference, upcoming)
- **Competition or Collusion? Entry Decisions in the Swedish Pharmaceutical Market**
  - European Commission DG-Comp (Brussels, 2026, invited seminar)
  - CEPR Health Economics Conference (Toulouse, 2026, conference)
  - European Association for Research in Industrial Economics EARIE (Mannheim, 2026, conference)
  - Competition and Regulation European Summer School and Conference CRESSE (Kos, 2026, conference)
  - Swedish Competition and Consumer Authority (online, 2026, invited seminar, upcoming)

Working Papers

- **Competition or Collusion? Entry Decisions in the Swedish Pharmaceutical Market**, with Otto Toivanen

  CEPR Discussion Paper DP21676, VoxEU Column

  Abstract: Price regulation may have unintended consequences, such as facilitating collusion or softening actual price competition by causing a reduction in the number of active firms. Observing different firms having a monopoly position in parts of a market may be the result of either intense post-entry competition and non-cooperative entry decisions, or collusive entry. This phenomenon and large price differences per pill within a market are widespread in the Swedish generic pharmaceutical market where price regulation through monthly auctions for each active ingredient – dosage form – strength – package size combination channels a large part of demand to the winner within each such substitution group. 22% of the markets are potentially collusive in having at least two firms as monopolies for different substitution groups at least some of the time and prices per pill in such markets are on average significantly higher. We take a structural model tailored to the Swedish circumstances to data on two markets. In one market, the entry patterns (monopoly package sizes) are suggestive of actual, in the other of attempted, collusion. Our counterfactual analysis where we induce substitution across package sizes yield savings of >50% in the first market, but lead to a modest expenditure increase in the other because intensified post-entry competition induces exit.

- **A Stylized Logit Demand Model with Internal Reference Pricing**

  Abstract: There is a growing literature applying logit demand models to pharmaceutical markets. However, some pharmaceutical markets differ from typical empirical IO markets, such as breakfast cereals, because of regulations including copayment structures and internal reference pricing. I extend the standard discrete-choice demand model to incorporate an internal reference price and a copayment structure. Using the Intermediate Value Theorem, I prove the existence of an equilibrium in a two-product market. The equilibrium is unique under additional restrictions: the two firms must be sufficiently different such that their best-response functions cross only once and do not overlap over

an interval. I then extend the existence result to an (n)-product market using an aggregative-game approach. The aim of this paper is twofold. First, I enrich the theory of logit demand models by introducing a stylized model that incorporates price regulation. Second, by establishing equilibrium existence and conditions for uniqueness, my model provides a theoretical foundation for empirical studies of pharmaceutical markets with internal reference pricing.

Works in Progress

- **Entry Decisions in the Swedish Pharmaceutical Market**, with Otto Toivanen

  We extend our analysis of the Swedish pharmaceutical market to study collusion in entry decisions across a broader set of markets beyond the two we have analyzed.

- **The Flow of Raw Materials in Finland**

  I construct a structural model to analyze how prices affect the flows of raw materials from production plants to manufacturing plants. Because transportation costs create geographically differentiated markets, firms could strategically set prices depending on the market power they have in a region.